

Update 4 – Weekly Update Report

CSE 499B: Senior Design II | Section: 12 | Group: 03

Week Update Summary

This week, we made good progress on multiple parts of the project. We merged the website features, fixed backend server issues on our local machine, resolved GitHub merge conflicts, and pushed the updated code to the main branch. We also set up separate virtual environments for the backend folders to make the project easier to run on other local machines, and created a fake job post website for testing the automation system. At the same time, we worked on the profile page by developing its front-end, back-end, and Supabase database integration, although we are still fixing some backend errors. We also made strong progress on the AI Mock Interview module by preparing and cleaning the dataset, integrating models like Qwen and FLAN-T5, implementing dynamic difficulty adjustment, fixing annotation-related errors, and testing the model's performance. The model works well with trained dataset answers, but its accuracy is lower on new outside answers, which shows that further improvement is still needed. Overall, this week was productive, and the remaining work on the profile page, resume checker, and mock interview improvements will continue in the next phase.

Contribution

Md. Kamrul Islam (2211745642):

This week, I have merged all the features of the website. During the merge, I got some issues running the backend server, and fixed them on my local machine. I built the virtual environment for all the backend server's folders separately so that it could be easier to run on any local machine. While pushing the merged code, I encountered conflicts between my current code and the main branch code. I fixed the conflicts and pushed the updated code to GitHub's main branch. I also created a fake job post website for testing the automation easily and hosting it online.

Farhan Ishraque (2212002042):

This week I worked on profile page and I worked on its front-end, backend and database. For front-end I used Javascript, react.js, for HTML like syntax inside Javascript JSX. I am working on its backend as it is giving me some error I intend to finish this work this week but I will in the next class. For the database I am using supabase database. I am working on it and thus my resume checker and profile will be ready in the next class.

Iram Shehzad (2131614642):

This week, I've made a lot of progress on my AI Mock Interview Module. I was able to set up the data processing, load and integrate the Qwen model, and put in place the main logic for adjusting the difficulty level dynamically. Everything is set up and working for an interview turn. I

got an `IndexError` in the cell-auto-annotate cell because I was trying to get elements from the strengths or weaknesses lists without first checking to see if they were empty. I addressed this by adding tests for the length of the list to make it stronger. It only makes one query at a time. I will run the auto-annotation again, and then I will go through the draft annotations by hand and make any necessary changes. After that, I'll get the data ready for fine-tuning, and then I'll move on.

Atikul Islam Nahid (2211978042):

This week, I worked on improving my mock interview project by training an AI model to give better answers. First, I set up the environment and installed all the required tools and libraries. Then I uploaded my dataset which contains software interview questions and answers. After that, I organized the dataset (like fixing category names and removing unnecessary spaces) so the model can learn properly. Next, I loaded model named (FLAN-T5) and prepared the data in a way that the model understands how to answer interview questions. I converted each question into a proper input format and paired it with the correct answer. Then I fine-tuned the model using this dataset. Mainly, I trained the model so it can generate good answers when given interview questions. Finally, I tested the system by running a mock interview where the model answers questions based on what it learned. I noticed that when I test the model using answers from my own dataset, the accuracy is high. But when I give answers from outside (new or different answers) the accuracy becomes lower

Completed Task:

- Various model training with the dataset.
- Read, write, and automatic Google Form submission.
- Merge all the features.
- Create a fake job webpage.

Work in Progress

- Comparing various models' performance.
- Integrating a custom model into the mock interview feature.
- Upload files to a Google Form and submit it automatically using the website.
- User profile data is fetched from the backend.

Next Target

- Integrate the custom model into the mock interview feature.
- Fully read and write on a Google Form and submit it automatically using the website.

Challenges

- Selecting and using the specific models for a specific task.
- Bypassing Google Chrome's automation blocking.
- Costly existing tools or websites in the current market.